

A CHAPTER IN PICTURES

The Turn That Changes Nothing

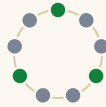
group theory introduced through al-Khalīl's circles of Arabic poetry



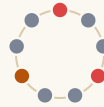
al-Mukhtalif



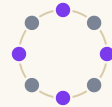
al-Mu'talif



al-Mujtalab



al-Mushtabih



al-Muttafiq

"Symmetry, as wide or as narrow as you may define its meaning, is one idea by which man through the ages has tried to comprehend and create order, beauty and perfection."

— HERMANN WEYL, *Symmetry* (1952)

§1 The Move You Cannot See

A ceiling fan has three identical blades. Turn it by exactly one third of a full circle. Each blade moves into its neighbour's position, and since the blades are identical, the fan looks untouched. A physical act left no visible trace.

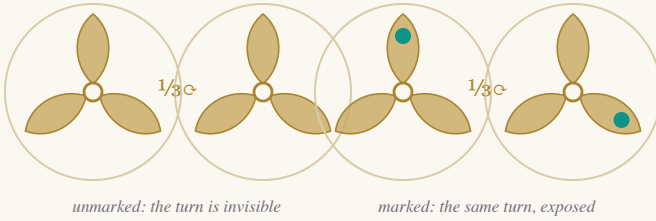


Fig. 1 — The same third-of-a-turn, twice. Unmarked (left), the move is invisible: this is what it means to be a symmetry. Mark one blade (right) and the same move becomes visible.

The definition:

A *symmetry* of a thing is a move that leaves it looking exactly as it was.

This defines symmetry as a move, not a property. It doesn't say the object is balanced or regular — it says: perform an action, and the object can't tell. A snowflake has sixth-turn symmetry. A square has quarter-turns and flips. A human face has roughly one mirror. To say what symmetry something has is to list the moves it fails to detect.

That shift — from an aesthetic quality to a set of operations — is the starting point of group theory.

§ 1 · CONTINUED

Three properties of these moves follow immediately.

First, doing nothing counts. Leaving the fan alone certainly leaves it unchanged. This is a symmetry, the same way zero is a number.

Second, every move has an inverse. If a third-turn clockwise is a symmetry, so is a third-turn counterclockwise.

Third, moves compose. Turn by a third, then by a third again: the result is a two-thirds turn, and the fan detects neither step. Combining symmetries always produces another symmetry.

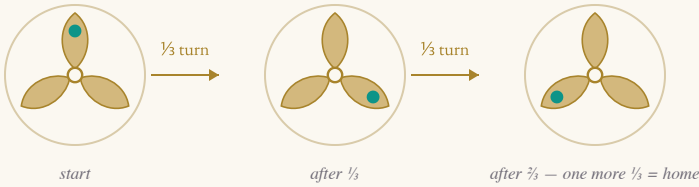


Fig. 2 — Moves compose. A third-turn followed by a third-turn is the two-thirds turn; a third-turn followed by a two-thirds turn returns every blade home — the do-nothing move.

These three properties — identity, inverses, closure under composition — define a **group**. Any set of moves satisfying them is one. The study of such sets is group theory.

The definition is minimal, and its reach is the surprise: it underlies modern physics, chemistry, and cryptography. The examples here are five circles drawn in eighth-century Basra to classify the metres of Arabic poetry.

§2 A Group, Properly Met

The fan has exactly three symmetries: *stay* (do nothing), a $\frac{1}{3}$ *turn*, and a $\frac{2}{3}$ *turn*. A quarter-turn is not one — it leaves the blades visibly askew. A full turn is just *stay* again.

Since moves compose, the whole system fits in a table: pick a row, follow it by a column, read off the result.

then → first ↓	stay	$\frac{1}{3}$ turn	$\frac{2}{3}$ turn
stay	stay	$\frac{1}{3}$ turn	$\frac{2}{3}$ turn
$\frac{1}{3}$ turn	$\frac{1}{3}$ turn	$\frac{2}{3}$ turn	stay
$\frac{2}{3}$ turn	$\frac{2}{3}$ turn	stay	$\frac{1}{3}$ turn

Fig. 3 — The composition table of the fan’s three symmetries. Every entry stays in the set; *stay* runs along the edges like a zero; the highlighted cells show each turn meeting its inverse.

The table certifies all four group axioms. Every entry stays in the set (*closure*). The *stay* row copies its header, like adding zero (*identity*). Each move has a partner that returns to *stay*: $\frac{1}{3}$ and $\frac{2}{3}$ undo each other, and *stay* undoes itself (*inverses*). And chains can be grouped in any order with the same result (*associativity*).

This group is the **cyclic group of order 3**, written C_3 . *Cyclic* because one move (the $\frac{1}{3}$ turn) generates all the others by repetition. *Order 3* because there are three moves.

§ 2 · CONTINUED

Clock arithmetic is a cyclic group. On a twelve-hour face, $9 + 5 = 2$: fourteen steps, but the wheel can't tell from two. That wrapping is C_{12} .

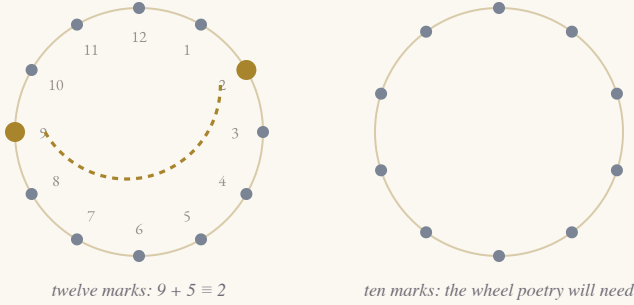


Fig. 4 — Two wheels, one idea. On twelve marks, advancing 9 then 5 lands on 2: the wheel forgets full revolutions. Any count that wraps — hours, weekdays, drumbeats — is run by a cyclic group C_n .

The general construction: place n marks on a wheel; the moves are *advance by k* for $k = 0, 1, \dots, n-1$; composition is addition mod n . This is C_n , the cyclic group of order n — the symmetry group of anything that repeats in a loop.

Cyclic groups are the simplest kind, which is why they work as an introduction. They require nothing beyond a wheel and modular addition. But they already exhibit the phenomena — actions, orbits, stabilizers, conservation laws — that govern much larger groups.

In the eighth century, al-Khalil ibn Aḥmad built five such wheels from the sounds of Arabic poetry and used them to answer a question: *how many genuinely different metres does the language permit?*

§3 The Poet's Raw Material

Arabic verse is quantitative: metre is a fixed pattern of long and short syllables. The tradition writes it with two marks. A **struck note** / is a consonant carrying a vowel. A **rest** 0 is an unvoiced consonant that closes a syllable. The marks read right-to-left, like the script.

These two symbols combine into a small set of units — the pegs and cords that make up every metre:

0/ light cord سبب خفيف strike, rest — a light two-step	0// joined peg وتد مجموع strike, strike, rest — the spine of most metres	/0/ split peg وتد مفروق strike, rest, strike — rare, and consequential (§8)	0/// little cluster فاصلة صغرى three strikes, rest — a quick drum-roll
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Fig. 5 — The four atomic units of Arabic prosody. Read each symbol from the right: 0// is strike–strike–rest. The colours follow these units through every figure.

Al-Khalīl ibn Aḥmad al-Farāhīdī (الخليل بن أحمد الفراهيدي)

Born in Oman around 718 CE, working in Basra until his death around 786, al-Khalīl compiled the first Arabic dictionary (*Kitāb al-ʿAyn*), organised by place of articulation; taught Sibawayh, author of the foundational Arabic grammar; and founded *ʿilm al-ʿarūd*, the science of metre. Tradition says the system came to him from the rhythm of a coppersmith's hammer in the market.

§ 3 · CONTINUED

The most common classical metre is الطويل (*aṭ-Ṭawīl*, "the long one"). Its half-line is فَعُولُنْ مَفَاعِيلُنْ فَعُولُنْ — in prosodic notation, the ten-unit sequence 0// 0/ 0// 0/ 0/ repeated twice. The same sequence returns line after line. The pattern repeats.

Al-Khalil's move was to close the loop. Join the end of the repeating sequence to its start, and the metre becomes a necklace of ten units.

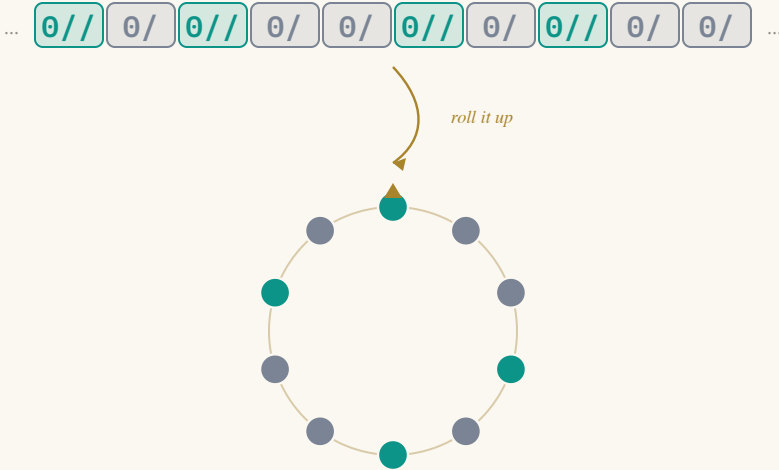


Fig. 6 — Rolling the repeating line into a circle. The tape's repetition becomes a single loop of ten units — teal pegs, grey cords. A mathematician calls this a quotient: the infinite line divided by its own repetition.

This is a quotient. Whenever a pattern repeats every n steps, positions that differ by n are equivalent. Identifying them turns an infinite line into a wheel with n marks — the wheel of Fig. 4. The circle is not a diagram of the metre; it *is* the metre, with repetition factored out.

Five such necklaces cover all of Arabic poetry. The first, دائرة المختلف (*the circle of the differing*), has ten beads: three pegs and seven cords. A wheel with ten marks carries a group: C_{10} .

§4 One Necklace, Many Clasps

A necklace has no beginning, but a metre must start somewhere. Choose a starting point — a *clasp* — and read around the loop. C_{10} , the ten rotations of a ten-bead wheel, *acts* on the necklace by moving the clasp from bead to bead.

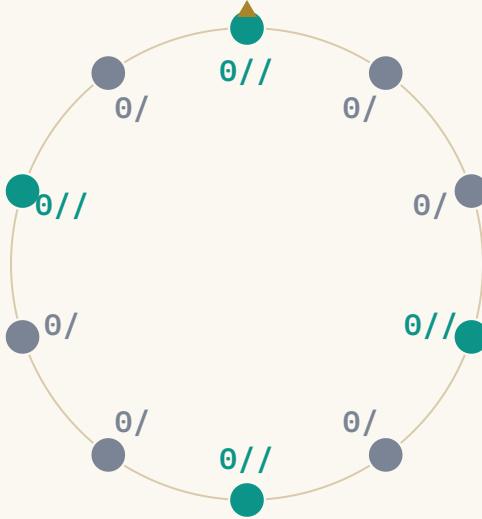


Fig. 7 — The circle of the differing (دائرة الاختلاف), beads labelled and coloured, read counter-clockwise from the marked clasp at the top. Reading from the top bead yields *فولن مضاعيلن* ... — the metre *aṭ-Ṭawīl*.

Move the clasp one bead forward and read again. The same beads, in the same cyclic order, yield a *different metre* — different feet, different rhythm, a different name in the tradition. Rotation changes what you hear.

When a group's elements operate on an object this way, the group *acts* on it. Each element gives a different view of the same object. In modern terms, al-Khalil's insight is: *the sixteen metres of Arabic poetry are not independent. They are five necklaces, read from different starting points under rotation.*

Ten rotations, ten possible clasps. The question is whether all ten yield distinct metres.

§ 4 · CONTINUED

Place the clasp at each bead in turn and read the pattern.

0	0//	0/	0//	0/	0/	0//	0/	0//	0/	0/	الطويل · al-Ṭawīl, "the long"
1	0/	0//	0/	0/	0//	0/	0//	0/	0/	0//	المديد · al-Madīd, "the protracted"
2	0//	0/	0/	0//	0/	0//	0/	0/	0//	0/	المستطيل · al-Mustaṭīl — possible, unused
3	0/	0/	0//	0/	0//	0/	0/	0//	0/	0//	البسيط · al-Basīṭ, "the outspread"
4	0/	0//	0/	0//	0/	0/	0//	0/	0//	0/	الممتد · al-Mumtadd — possible, unused
5– 9	the same five readings again, in the same order — nothing new appears										

Fig. 8 — All ten clasp positions of the first circle. Five distinct readings emerge — three canonical metres and two lawful but unused ones — and then the list repeats.

Two features of this list need explaining.

The list is short. Ten rotations produced five distinct readings. Position 5 sounds exactly like position 0, position 6 like position 1. Some rotations produce no new reading. Why five? That is the subject of §5 and §6.

The list has gaps. Poets use three of the five: الطويل، المديد، البسيط. The other two are well-formed rhythms nobody adopted. Al-Khalīl named them rather than discarding them. §9 returns to these empty cells.

§5 The Orbit: What Is Genuinely Different

The set of distinct readings is what group theory calls the **orbit** of the necklace under rotation: every result the group’s moves produce, counted without repetition.

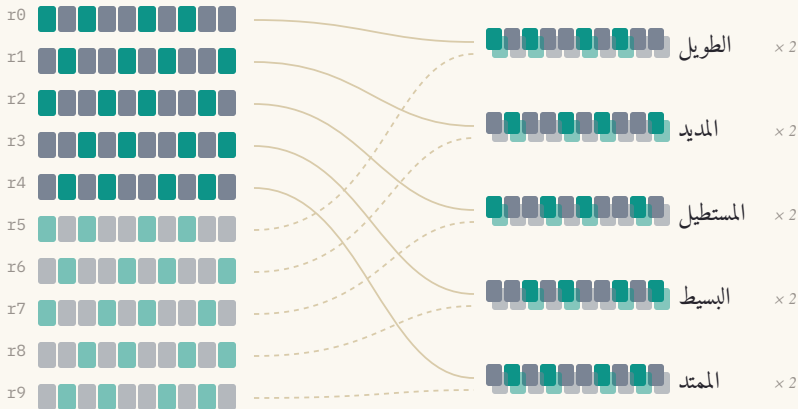


Fig. 9 — The ten rotations of circle 1, dealt into piles that sound identical. Ten readings collapse into five piles of exactly two. The orbit has five members; the pile-height — here, two — gets its own name in §6.

The orbit is the actual yield. The group has ten moves, but only five produce distinct results. The same count applies to all five of al-Khalil’s circles:

CIRCLE	BEADS N	ORBIT (DISTINCT READINGS)
المختلف · the differing	10	5
المؤتلف · the concordant	6	2
المجتلب · the brought-in	9	3
المشتبه · the resembling	9	9
المتفق · the agreeing	8	2

Fig. 10 — Five necklaces, audited. The orbits sum to $5+2+3+9+2 = 21$ distinct readings across the system. The two nine-bead circles have very different yields: 3 against 9. §8 explains why.

§6 The Stabilizer: The Necklace's Own Symmetry

Why does position 5 sound the same as position 0? The first circle's sequence is 0// 0/ 0// 0/ 0/ — repeated twice. Two identical halves. Rotating by five beads sends every peg to a peg and every cord to a cord. The necklace can't tell the difference. This is the fan's symmetry, applied to beads.

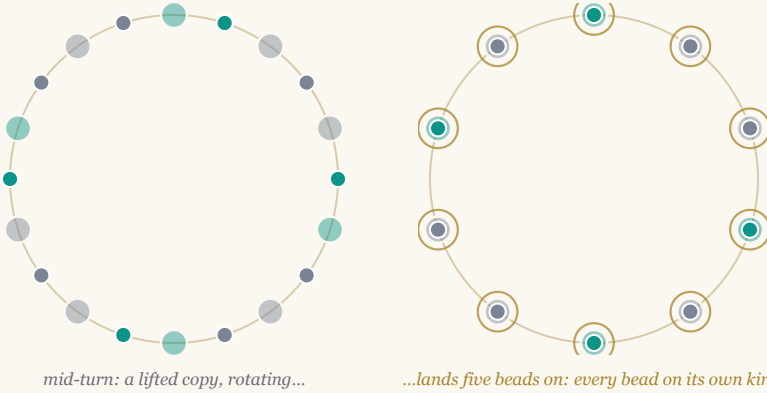


Fig. 11 — The turn that changes nothing, performed. A ghost copy of the necklace rotates by five positions and settles exactly onto the original — peg on peg, cord on cord.

The moves that leave a necklace unchanged form a subgroup — they compose and invert within themselves. This subgroup is the **stabilizer**. For circle 1 the stabilizer is { stay, turn-by-5 }: C_2 .

This explains why the list is short. Applying a stabilizer move to any clasp position leaves the reading unchanged. Ten positions, paired by the stabilizer, give five distinct readings. More symmetry means fewer distinct readings.

§ 6 · CONTINUED

The stabilizer can be drawn. Connect each bead to the bead the stabilizer sends it to. The chords form a star polygon, and because the move preserves bead types, every chord joins two beads of the same colour.

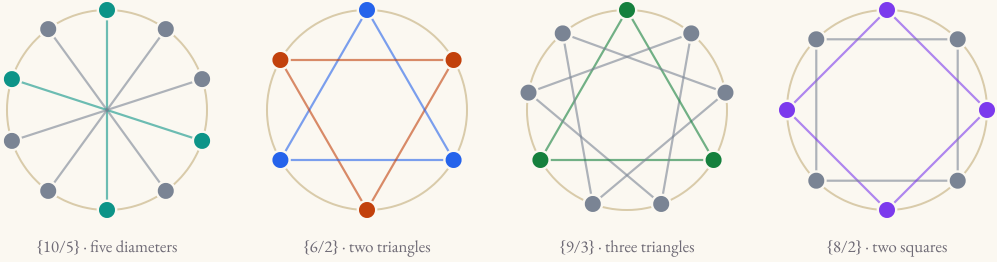


Fig. 12 — Four circles, their stabilizers drawn as stars. In the concordant circle the hexagram splits into a blue peg-triangle and a rust cluster-triangle: the symmetry sorts the beads by kind. One circle is missing — it draws no star at all, and §8 explains why.

The shepherd's theorem

Forty legs, four per sheep, ten sheep. Divide a total by a uniform redundancy to get a count. This is, almost literally, the theorem on the next page.

One distinction worth making: the group C_{10} belongs to the wheel — the ten positions. The stabilizer belongs to the *beading*: thread the same wheel with different beads and the stabilizer changes. Symmetry is a property of the pattern, not the positions.

§7 A Conservation Law

Three counts on the first circle: the group has 10 moves, the orbit has 5 readings, the stabilizer has 2 elements.

$$10 = 5 \times 2$$

$$\text{moves} = \text{distinct readings} \times \text{symmetries}$$

This is not a coincidence. Sort the ten rotations by which reading they produce. Each pile has exactly the stabilizer's size: two rotations that give the same reading differ by a stabilizer element. Every pile is a copy of the stabilizer, shifted to a different starting point.



Fig. 13 — The ten moves of C_{10} , dealt by the reading they produce. Five piles, each of height exactly two — the stabilizer's size. Equal piles are the whole proof: $10 = 5 \text{ piles} \times 2 \text{ per pile}$.

In full generality, this is the **orbit–stabilizer theorem**: for a finite group acting on an object,

$$\text{size of group} = \text{size of orbit} \times \text{size of stabilizer}.$$

The product is fixed, so symmetry and variety trade off. More symmetry means fewer distinct readings, and vice versa. The group size is the budget; orbit and stabilizer split it.

§7 · CONTINUED

All five circles, each row an independent check:

CIRCLE	BEADS	ORBIT	STABILIZER	THE LAW
المختلف	10	5	C_2	$10 = 5 \times 2$
المؤتلف	6	2	C_3	$6 = 2 \times 3$
المجتلب	9	3	C_3	$9 = 3 \times 3$
المشتبه	9	9	C_1	$9 = 9 \times 1$
المتفق	8	2	C_4	$8 = 2 \times 4$

Fig. 14 — One theorem, five instances. Every circle satisfies beads = orbit $\times$ stabilizer exactly. (All counts verified by exhaustive computation; see the colophon.)

Read downward, this is five instances of one multiplication. Read across, it is a trade-off. The concordant and agreeing circles have the largest stabilizers (C_3 and C_4) and produce the fewest readings — two each. The resembling circle has no symmetry beyond the identity and produces the most: all nine rotations are distinct.

The two middle rows are the interesting case. Both necklaces have nine beads and the same group C_9 . One yields three readings, the other nine. Same count, different structure. §8 explains the difference.

§8 Symmetry Comes in Steps, Not Slopes

Before comparing the two nine-bead circles: the stabilizer's size must divide the group's size exactly. Equal piles divide the total with no remainder. (This is **Lagrange's theorem**, which holds for every subgroup of every finite group.) The consequence: possible symmetry sizes are not continuous. They are a short list of divisors.

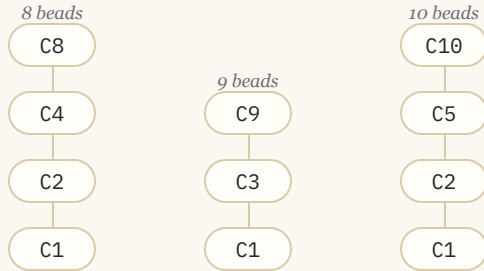


Fig. 15 — *The only symmetries a necklace may have, by Lagrange: divisor ladders. On nine beads nothing exists between C_9 and C_1 — there is no intermediate symmetry. (The top rung, all beads identical, occurs in no real circle.)*

On nine beads, the options are C_9 (only if all beads are identical), C_3 , or C_1 . Nothing in between. A nine-bead necklace that lacks C_3 symmetry doesn't degrade gracefully — it drops directly to C_1 , the trivial group. There is no intermediate step.

The two nine-bead circles sit on opposite sides of this gap.

§ 8 · CONTINUED

Compare the two nine-bead circles bead by bead:

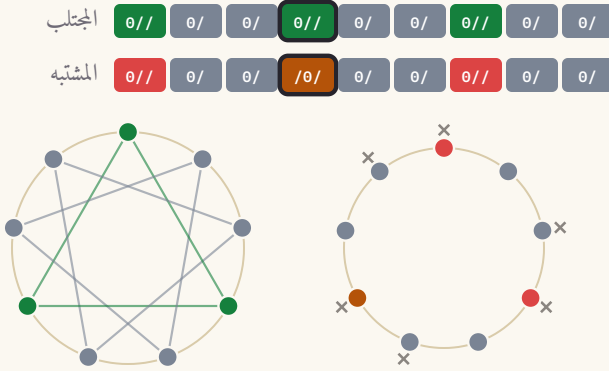


Fig. 16 — One bead of difference. The brought-in circle repeats peg–cord–cord three times: symmetry C_3 , three readings. The resembling circle replaces the middle peg with the rare split peg $/0/$ — and the smallest turn now sets six beads on unlike beads (the $\times$ marks). Symmetry: none. Readings: all nine.

One bead changed out of nine — the smallest possible alteration. By the divisor ladder, the result is total: C_3 drops to C_1 , orbit 3 becomes orbit 9. The two most similar circles in the system have the greatest difference in symmetry.

Al-Khalil’s tradition named this circle **المشبه** — *the resembling*, and simultaneously *the confusable*: in Arabic the word carries both senses. Its nine rotations are near-misses of one another, maximally similar while remaining distinct. (The best-known confusions in practice come from a different source — lossy metrical variations that let one circle’s metre sound like another’s. That is a separate mechanism, covered in the companion chapter on **تداخل**.)

A note on invariants: the stabilizer records *that* symmetry is absent — C_1 , the same as random noise. It doesn’t record how close the necklace came to being symmetric. The group-theoretic invariant loses that information. The traditional name **المشبه** preserves it.

§9 The Catalogue and Its Gaps

The orbits sum to $5 + 2 + 3 + 9 + 2 = 21$ distinct readings across all five circles. Classical poetry uses sixteen. Al-Khalil kept all twenty-one.

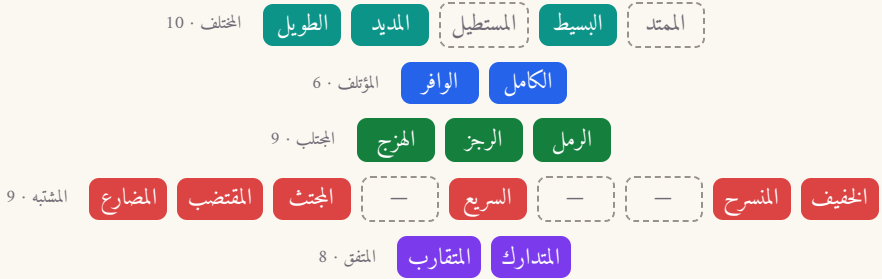


Fig. 17 — The complete theoretical catalogue: 21 cells, one per distinct reading, row by circle. Sixteen are inhabited by the classical metres. Five are empty (dashed) — well-formed rhythms no poet adopted. Al-Khalil named the first circle's two vacancies; the resembling circle's three were left nameless.

A catalogue built from evidence lists what occurs. A catalogue built from a group action lists what is *possible*. The orbit doesn't depend on attestation. The five vacancies are structural predictions: rhythms the system permits, independent of what poets wrote. Absence becomes a fact requiring explanation.

Mendeleev did the same with elements: trust the structure, leave the cells empty, let experiment fill them. Al-Khalil's empty cells remain empty — poets are not obliged — but the vacant metres were tried, and المستطيل and الممتد survive in the manuals as named but unused readings. All five gaps fall in the two least symmetric circles, the only ones whose orbits exceeded what the tradition used.

§10 The Price of Symmetry

The orbit–stabilizer theorem fixes the product *orbit* $\times$ *stabilizer*. Across the five circles, the trade-off is visible:

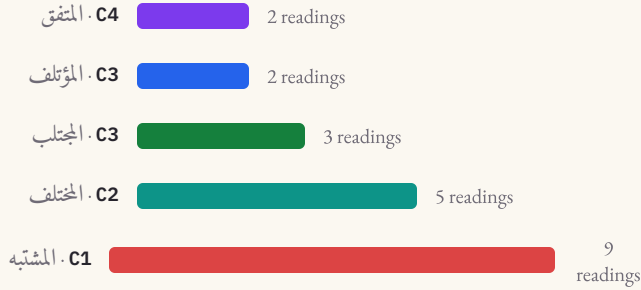


Fig. 18 — The inverse law, drawn. Circles sorted by symmetry, from the most symmetric (C_4) to the least (C_1): the yield of distinct readings runs the other way.

In ordinary language, symmetry is a good thing. In group theory, symmetry is sameness, and sameness reduces variety. Every undetectable turn is a clasp position that yields no new reading. Perfect symmetry would mean one reading — monotony. Distinct things come from broken symmetry.

The same trade-off appears at other scales. The early universe cooled through broken symmetries into the particle zoo. A crystal’s lattice hosts one pattern; a disordered glass hosts many. Al-Khalil’s circles show it in nine beads: the concordant circle (C_3) produces two metres; the resembling circle (C_1) produces nine.

§10 · CLOSING

The components assembled here: a set of objects (rhythms); a group acting on them (rotations); an equivalence relation (same reading); the equivalence classes counted (orbits = metres); each object's symmetry measured (stabilizer); a conservation law (orbit–stabilizer); a complete catalogue derived from structure, not evidence; and predicted absences. This is the classification method of modern mathematics. Al-Khalil executed every step, by ear, eleven centuries before Galois named the subject.

What the circles show about group theory is that it is not a notation. It is a framework for answering a question: *when are two things genuinely the same, and how many genuinely different things exist?* Ask that question carefully enough — of molecules, ciphers, or verse — and you will find orbits and stabilizers, whether or not you have names for them. The formalism didn't create the symmetry of poetry. It gave a name to what a lexicographer had already found.

Hear it for yourself

Every figure here is a still from the *Interactive Arud Explorer* at arudi.midadalfikr.com. The interactive version lets you rotate each circle's dial and watch metres change; perform the ghost-rotation of Fig. 11; draw the star polygons of Fig. 12; collapse rotations into orbit piles; and play any necklace as a drum pattern while moving the clasp.

When you next turn something and no one notices — a fan, a bolt, a phrase — you have performed a stabilizer element.

Glossary & Notes

symmetry

a move that leaves a thing looking exactly as it was (§1).

group

a set of moves containing *do-nothing*, closed under composition, with every move invertible (§2).

cyclic group C_n

the n rotations of an n -mark wheel; clock arithmetic (§2).

action

a group applied to an object, each move giving a different view of it (§4).

orbit

every result an object takes under the group's moves, without repetition — here, the distinct metres of a circle (§5).

stabilizer

the moves an object cannot detect; its symmetry subgroup (§6).

orbit–stabilizer theorem

group size = orbit $\times$ stabilizer (§7).

Lagrange's theorem

a subgroup's size divides the group's size; symmetry comes in divisor steps (§8).

quotient

identifying positions declared equivalent — rolling the repeating line into the circle (§3).

عروض 'arūd

the science of Arabic metre, founded by al-Khalil (§3).

sabab سبب, watid وتد, fāṣila فاصلة

cord, peg, and cluster — the atomic rhythm units (Fig. 5).

muhmal مَهْمَل

"neglected": a lawful reading no poet adopted — the catalogue's empty cells (§9).

the five circles

المتخلف the differing (10 beads), المؤتلف the concordant (6), المتجلبب the brought-in (9), المشتبّه the resembling (9), المتفق the agreeing (8).

the sixteen metres

الطويل، المديد، البسيط · الوافر، الكامل · الخرج، الرجز، الرمل · السريع، المنسرح، الخفيف، المضارع، المقتضب، المجتث · المتقارب، المتدارك.

FURTHER READING. H. Weyl, *Symmetry* (1952) — the classic meditation; M. A. Armstrong, *Groups and Symmetry* (1988) — a first formal course; and any classical manual of *'ilm al-'arūd* for the circles as the tradition drew them.

COLOPHON. Set in EB Garamond, Cormorant Garamond, Amiri, and IBM Plex Mono. All counts used in this chapter — orbits 5, 2, 3, 9, 2; stabilizers C_2 , C_3 , C_3 , C_1 , C_4 ; $21 = 16 + 5$ — were verified by exhaustive computation against the sequence data of the Interactive Arud Explorer. Prosodic symbols appear in classical right-to-left order throughout: 0// is read strike–strike–rest, from the right.